

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-2500: Engineering Probability, Spring 2026
Final Exam Solutions

The overall mean was a 68. There were 2 perfect scores and 8 scores in the 90's.

1. (12 points.) There are 4 player classes in Slay the Spire 2, which have different starting healths and varied popularity among players. The probabilities that a player chooses each class are given in the table below.

Class	Starting health	Probability
Ironclad	80	0.4
Defect	75	0.2
Silent	70	0.3
Regent	60	0.1

- (a) (6 points.) Let H be the starting health of a randomly chosen player. Compute $E(H)$ and $\text{Var}(H)$.

We compute these values directly using the definition of expected value:

$$\begin{aligned} E(H) &= 80(0.4) + 75(0.2) + 70(0.3) + 60(0.1) &= 74 \\ \text{Var}(H) &= (80 - 74)^2(0.4) + (75 - 74)^2(0.2) + (70 - 74)^2(0.3) + (60 - 74)^2(0.1) &= 39 \end{aligned}$$

- (b) (2 points.) Suppose a group of players has probability $\frac{1}{5}$ of winning the game on any given attempt, and let N be the number of attempts the group needs to win. What kind of random variable is N ?

This is a geometric random variable (number of trials until success).

- (c) (4 points.) Suppose the length required for a single attempt is a (continuous) uniform random variable on $[1, 2]$ hours. Using this and the information in part (b), determine the expected length of time required to win the game once.

From part (b), the expected number of attempts to win is $\frac{1}{1/5} = 5$. From part (c), the expected length of each attempt is 1.5 hours. So the expected time to the first win is $5 \cdot 1.5 = 7.5$ hours.

Yes, I know about the Necrobinder.

2. (16 points.) While playing Overcooked, orders for burgers can have **0, 1, or 2** of the 3 possible toppings cheese, lettuce, and tomato. (A burger that has all 3 toppings is not an option.)

- (a) (3 points.) What is the sample space for the toppings on a burger? How many outcomes are there?

Letting C = cheese, L = lettuce, T = tomato, and $\emptyset$ = no toppings, the sample space is

$$\{\emptyset, C, L, T, CL, CT, LT\}$$

There are 7 outcomes in the sample space. You had to clearly enumerate the sample space to get full credit (just saying $1 + 3 + 3$ is not defining the sample space).

- (b) (5 points.) Assuming each possibility in part (a) is equally likely, compute the probability that exactly 3 of the first 5 orders contain lettuce.

This is like a binomial random variable where success = {contains lettuce}, which has probability $\frac{3}{7}$. So the probability is

$$\binom{5}{3} \left(\frac{3}{7}\right)^3 \left(\frac{4}{7}\right)^2 = 10 \cdot \frac{432}{16807} = \frac{4320}{16807} = 0.257$$

- (c) (8 points.) Assuming each possibility in part (a) is equally likely, compute the probability that at least 2 orders containing lettuce arrive before the first order containing tomato. Note: you may need the infinite sum formula $\sum_{n=1}^{\infty} n a^n = \frac{a}{(1-a)^2}$.

The easiest approach is to compute

$$P(\geq 2 L \text{ before } T) = 1 - P(0 L \text{ before } T) - P(1 L \text{ before } T)$$

What is $P(0 L \text{ before } T)$? This means that we can get as many orders in $\{\emptyset, C\}$ as we want (including 0 of these orders), and then one of the orders in $\{T, CT, LT\}$. This probability is

$$\begin{aligned} P(0 L \text{ before } T) &= \left(\frac{3}{7}\right) + \left(\frac{2}{7}\right)\left(\frac{3}{7}\right) + \left(\frac{2}{7}\right)^2\left(\frac{3}{7}\right) + \dots \\ &= \left(\frac{3}{7}\right) \sum_{k=0}^{\infty} \left(\frac{2}{7}\right)^k \\ &= \left(\frac{3}{7}\right) \frac{1}{1 - \frac{2}{7}} \\ &= \frac{3}{7} \cdot \frac{7}{5} \\ &= \frac{3}{5} \end{aligned}$$

The other term means that we can have exactly one order in $\{L, CL\}$ before the T order. If there are k orders, then the L-but-not-T order can occur in any of those k positions (i.e., $\binom{k}{1} = k$). So this probability is

$$\begin{aligned}P(1 \text{ L before } T) &= \left(\frac{2}{7}\right)\left(\frac{3}{7}\right) + 2 \cdot \left(\frac{2}{7}\right)^2 \left(\frac{3}{7}\right) + 3 \cdot \left(\frac{2}{7}\right)^3 \left(\frac{3}{7}\right) + \dots \\&= \left(\frac{3}{7}\right) \sum_{k=0}^{\infty} k \left(\frac{2}{7}\right)^k \\&= \left(\frac{3}{7}\right) \frac{\frac{2}{7}}{\left(1 - \frac{2}{7}\right)^2} \\&= \frac{3}{7} \cdot \frac{2}{7} \cdot \frac{49}{25} \\&= \frac{6}{25}\end{aligned}$$

Thus the original probability we want is

$$1 = \frac{3}{5} - \frac{6}{25} = \frac{4}{25} = 0.16$$

Surprisingly, there were some incorrect ways of getting the same answer, but I needed to be convinced that you understood the underlying logic to get full credit.

3. (16 points.) Grogg enjoys using the Force, but it also makes him sleepy. Let X be the length in minutes that he uses the Force, and Y be the length in minutes of the resulting nap. Suppose X is a (continuous) uniform random variable on $[0, 10]$. The conditional PDF of Y given X , $f_{Y|X}(y|x)$, is a (continuous) uniform random variable on $[\frac{1}{2}X^2, \frac{3}{2}X^2]$.

- (a) (5 points.) Using the law of iterated expectations, compute the expected value of Y .

The law of iterated expectations says

$$\begin{aligned} E(Y) &= E(E(Y | X)) \\ &= E(X^2) \\ &= \text{Var}(X) + (E(X))^2 \\ &= \frac{100}{12} + 5^2 \\ &= \frac{100}{3} \end{aligned}$$

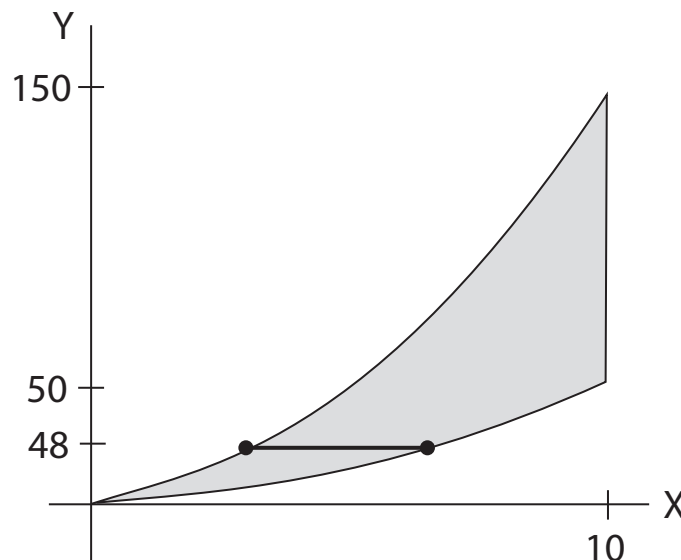
Here, the second line comes from the fact that the distribution of Y given X is uniform on $[\frac{1}{2}X^2, \frac{3}{2}X^2]$ so its mean is the midpoint of the interval, or X^2 . The fourth line comes from the known mean and variance of X , a uniform random variable on $[0, 10]$.

- (b) (5 points.) Determine the formula for the joint PDF $f_{XY}(x, y)$. This **must** include a sketch of the region in the (X, Y) plane where the joint PDF is greater than 0. Be sure to label all important points on the X and Y axes.

The joint is the marginal times the conditional, i.e.,

$$\begin{aligned} f_{XY}(x, y) &= f_X(x) \cdot f_{Y|X}(y|x) \\ &= \frac{1}{10-0} \frac{1}{\frac{3}{2}x^2 - \frac{1}{2}x^2} \\ &= \frac{1}{10x^2} \end{aligned}$$

This is only valid in the shaded region pictured below, and is 0 elsewhere.



- (c) (6 points.) Determine the value of the marginal PDF $f_Y(y)$ at the value $y = 48$. Referring to your sketch in (b) may help. You only need to compute the value of the PDF at this one point!

We only need to compute the marginal for the value $y = 48$, which we get by integrating the joint along the line indicated in the picture above. We can compute that the left and right endpoints of the interval correspond to $\frac{x^2}{2} = 48 \Rightarrow x = \sqrt{96}$ and $\frac{3x^2}{2} = 48 \Rightarrow x = \sqrt{32}$. So the integral is:

$$\begin{aligned} f_Y(48) &= \int_{-\infty}^{\infty} f_{XY}(x, 48) dx \\ &= \int_{\sqrt{32}}^{\sqrt{96}} \frac{1}{10x^2} dx \\ &= \left. \frac{-1}{10x} \right|_{\sqrt{32}}^{\sqrt{96}} \\ &= \frac{1}{10\sqrt{32}} - \frac{1}{10\sqrt{96}} \\ &= \frac{1}{40\sqrt{2}} \left(1 - \frac{1}{\sqrt{3}} \right) \\ &= 0.0075 \end{aligned}$$

4. (10 points.) The midichlorian count of a normal person has a Gaussian distribution with mean 4 and variance 1. The midichlorian count of a Force-sensitive person has a Gaussian distribution with mean 16 and variance 1.

The Jedi Council uses a pooled testing scheme to identify Force-sensitive candidates. Their method is to collect samples from 4 candidates (all of which are independent) and pour them all together so that the midichlorian count S of the resulting sample is the sum of the four individual midichlorian counts. Let H_0 be the hypothesis that there are no Force-sensitive candidates in the pool, with prior probability 0.9999. Let H_1 be the hypothesis that there is exactly one Force-sensitive candidate in the pool, with prior probability 0.0001. (The possibility of having 2 or more Force-sensitive candidates in the pool is so small we don't even consider it.)

- (a) (4 points.) Determine the class-conditional distributions $f_{S|H_0}(x)$ and $f_{S|H_1}(x)$. Be as complete as you can (i.e., an explicit formula for each PDF).

Under H_0 , we have the sum of four $N(4, 1)$ distributions, which is itself a Gaussian with mean 16 and variance 4. Thus its PDF is:

$$f_{H_0}(x) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(x-16)^2}{8}\right)$$

Under H_1 we have the sum of three $N(4, 1)$ distributions and one $N(16, 1)$ distribution, which is also a Gaussian with mean 28 and variance 4. Thus its PDF is:

$$f_{H_1}(x) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(x-28)^2}{8}\right)$$

- (b) (6 points.) Suppose a pool is tested and the result comes back $S = 25$. Which hypothesis should we choose if we apply the Maximum A Posteriori decision rule?

First we compute the likelihood ratio

$$\begin{aligned} \frac{f_{H_0}(x)}{f_{H_1}(x)} &= \left(\frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(x-16)^2}{8}\right) \right) / \left(\frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(x-28)^2}{8}\right) \right) \\ &= \exp\left(\frac{-(x-16)^2}{8} + \frac{(x-28)^2}{8}\right) \end{aligned}$$

At the given value of $X = 25$ we get

$$\begin{aligned} \exp\left(\frac{-(9)^2}{8} + \frac{(-3)^2}{8}\right) &= \exp\left(\frac{-72}{8}\right) \\ &= \exp(-9) \\ &= 0.000123 \end{aligned}$$

Now we compare this to the ratio of prior probabilities

$$\frac{P(H_1)}{P(H_0)} = \frac{0.0001}{0.9999} = 0.00010001$$

Since the likelihood ratio is bigger than this value we should conclude that this is a normal batch (i.e., no Force-sensitive users in the pool).

You may have gotten some partial credit if your answer was wrong in (a), but some wrong answers make the problem trivial (e.g., if you mistakenly have the variances as 1, then your numbers will probably be way too large or way too small, which I've told you to look out for as the sign of a wrong answer).

5. (12 points.)

- (a) (6 points.) A vampire is low on health (20 hit points out of a possible 100). The i^{th} person he bites gives him an amount of health X_i , an exponential random variable with parameter $\lambda = \frac{1}{3}$. All of the X_i are independent. Using the Central Limit Theorem, determine the probability that he must bite more than 20 people to regain full health.

The statement is equivalent to computing $P(S_{20} < 80)$, where S_{20} is the sum of 20 bites. X_i is exponential, so $\mu = 3$ and $\sigma = 3$. The Central Limit Theorem says

$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \geq z\right) \approx Q(z)$$

or

$$P(S_n < n\mu + z\sigma\sqrt{n}) \approx 1 - Q(z)$$

Here $n = 20$ so

$$20(3) + z(3)(\sqrt{20}) = 80$$

Thus

$$z = \frac{80 - 60}{3\sqrt{20}} = 1.49$$

So the probability is about $1 - Q(1.5) = 0.9332$.

- (b) (6 points.) Instead of biting people, the vampire tries biting pandas. The i^{th} panda he bites gives him an amount of health Y_i , which comes from a Gaussian distribution with unknown mean μ and variance 4. After biting 15 pandas, the vampire finds his health has increased by 72. Find the 95% confidence interval for μ .

The confidence interval for the mean is

$$M_{15} \pm z_{0.025} \frac{\sigma}{\sqrt{15}}$$

From the Q table, $z_{0.025} = 1.96$. We are told $\sigma = 2$, and our sample mean $M_{15} = \frac{72}{15} = 4.8$. So the 95% confidence interval is

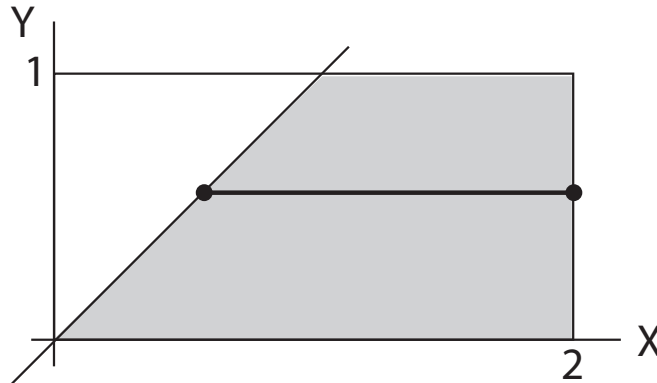
$$4.8 \pm \frac{(1.96)(2)}{\sqrt{15}} = 4.8 \pm 1.01 = [3.79, 5.81]$$

6. (14 points.) Consider the pair of random variables X and Y whose joint PDF is given by

$$f_{XY}(x, y) = \begin{cases} \frac{1}{3}(3 - x - y) & x \in [0, 2], y \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

- (a) (6 points.) Compute $P(X > Y)$.

The region of interest is sketched at right. One way to do it is to integrate along horizontal lines as indicated in the figure, giving:



$$\begin{aligned} P(X > Y) &= \frac{1}{3} \int_0^1 \int_y^2 3 - x - y \, dx \, dy \\ &= \frac{1}{3} \int_0^1 \left(3x - \frac{1}{2}x^2 - xy \right) \Big|_{x=y}^{x=2} dy \\ &= \frac{1}{3} \int_0^1 4 - 5y + \frac{3}{2}y^2 \, dy \\ &= \frac{1}{3} \left(4y - \frac{5}{2}y^2 + \frac{1}{2}y^3 \right) \Big|_{y=0}^{y=1} \\ &= \frac{1}{3}(2) \\ &= \frac{2}{3} \end{aligned}$$

- (b) (4 points.) Given that $E(X) = \frac{7}{9}$ and $E(Y) = \frac{4}{9}$, compute $\text{Cov}(X, Y)$. Be sure to show your work (i.e., set up and manually do the necessary integral).

The covariance of X and Y is defined as $E(XY) - E(X)E(Y)$. We are given the last two values already, so we just have to compute $E(XY)$.

$$\begin{aligned}
E(XY) &= \frac{1}{3} \int_0^1 \int_0^2 3xy - x^2y - xy^2 \, dx \, dy \\
&= \frac{1}{3} \int_0^1 \left(\frac{3}{2}x^2y - \frac{1}{2}x^2y^2 - \frac{1}{2}x^2y^2 \right) \Big|_{x=0}^{x=2} dy \\
&= \frac{1}{3} \int_0^1 \frac{10}{3}y - 2y^2 \, dy \\
&= \frac{1}{3} \left(\frac{5}{3}y^2 - \frac{2}{3}y^3 \right) \Big|_{y=0}^{y=1} \\
&= \frac{1}{3} \cdot 1 \\
&= \frac{1}{3}
\end{aligned}$$

Thus

$$\text{Cov}(X, Y) = \frac{1}{3} - \frac{7}{9} \cdot \frac{4}{9} = \frac{-1}{81}$$

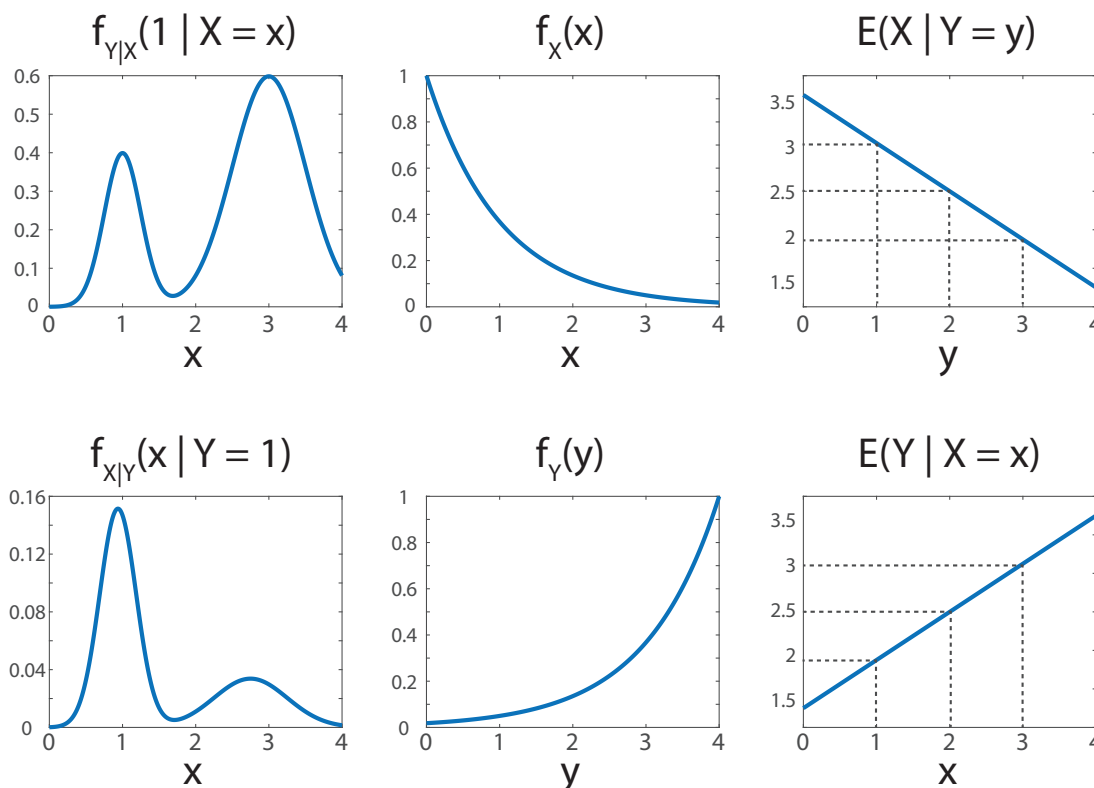
- (c) (2 points.) Is the correlation coefficient ρ positive, negative, or 0? You should be able to explain your answer without actually computing the value of ρ .

Since the correlation coefficient is just the covariance divided by a positive number, we can directly conclude the correlation coefficient is negative.

- (d) (2 points.) Are X and Y independent? Explain your answer.

Since the variables are correlated, they are not independent.

7. (9 points.)



The graphs above illustrate various functions derived from the (not given) joint PDF of two random variables X and Y , each of which is defined in the range $[0, 4]$. Using only this information (i.e., without any extra computation), determine the following values. In each case, indicate which graph(s) you're basing your answer on, and give a short explanation of your reasoning. You might not need to use every graph to answer these questions.

(a) (3 points.) The maximum likelihood (ML) estimate of X given that $Y = 1$.

The ML estimate is the value of X that maximizes the conditional probability $f_{Y|X}(y|x)$, which we can see from the upper left picture is $X = 3$.

Aside from mixing up the ML and MAP estimates, another common mistake was to return the height of the peak instead of the value of X where that peak occurs (this is the difference between max and arg max). Also some people weren't clear that a numerical answer was expected in each part of the problem.

(b) (3 points.) The maximum a posteriori (MAP) estimate of X given that $Y = 1$.

The MAP estimate is the value of X that maximizes the conditional probability $f_{X|Y}(x|y)$, which we can see from the lower left picture is $X = 1$.

(c) (3 points.) The minimum mean-squared error (MMSE) estimate of X given that $Y = 1$.

The MMSE estimate is the conditional expected value $E(X|Y = 1)$, which we can see from the upper right picture is $X = 3$.

8. (11 points.) When a burrito is ordered, the chef needs to boil some rice and cook some beans. The length of time R it takes to make the rice is an exponential random variable with mean 5 minutes, and the length of time B it takes to make the beans is an exponential random variable with mean 6 minutes. B and R are independent random variables. The burrito is ready when both the rice and beans are done, so whichever one takes longer is the limiting factor.

- (a) (5 points.) Determine the CDF of the preparation time $P = \max(B, R)$.

Proceeding from first principles,

$$\begin{aligned}
 P(P \leq x) &= P(\max(B, R) \leq x) \\
 &= P(B \leq x \text{ and } R \leq x) \\
 &= P(B \leq x) \cdot P(R \leq x) \\
 &= \left(1 - \exp\left(-\frac{1}{6}x\right)\right) \left(1 - \exp\left(-\frac{1}{5}x\right)\right) \\
 &= 1 - \exp\left(-\frac{1}{6}x\right) - \exp\left(-\frac{1}{5}x\right) + \exp\left(-\frac{11}{30}x\right)
 \end{aligned}$$

This holds for $x \geq 0$, and is 0 otherwise.

There were many major misconceptions on this problem (e.g., just choosing the PDF that had the bigger value, making an if statement based on the crossover point of the two PDFs, or assuming that the beans will always take longer than the rice). Also, some people jumped right to multiplying the CDFs, but for full credit you need to provide a chain of reasoning to show that you understand why this works (for example, what would you have done if we had instead defined $P = \min(B, R)$?). To get full credit I needed something that clearly indicated that you understood both R and B had to be less than a target value, and that since these were independent you can multiply the two CDFs.

- (b) (3 points.) Determine the PDF of P .

The PDF is the derivative of the CDF:

$$\begin{aligned}
 f_P(x) &= \frac{d}{dx} \left(1 - \exp\left(-\frac{1}{6}x\right) - \exp\left(-\frac{1}{5}x\right) + \exp\left(-\frac{11}{30}x\right)\right) \\
 &= \frac{1}{6} \exp\left(-\frac{1}{6}x\right) + \frac{1}{5} \exp\left(-\frac{1}{5}x\right) - \frac{11}{30} \exp\left(-\frac{11}{30}x\right)
 \end{aligned}$$

This holds for $x \geq 0$, and is 0 otherwise.

You got some credit if you correctly took the derivative of an incorrect answer in (a) to get (b). However, if you made the problem too easy by mistakenly arriving at a simple exponential PDF in (a), you won't get full credit for parts (b) and (c).

- (c) (3 points.) Compute $E(P)$. Note that you should be able to determine this quickly from your answer in (b) without much computation.

We can see that the PDF is the weighted sum of three exponential PDFs and the weights add up to 1, so we can quickly conclude that the expected value is the corresponding weighted sum of the means without having to do any extra computations:

$$\begin{aligned}
 E(P) &= 5 + 6 - \frac{30}{11} \\
 &= \frac{91}{11} = 8.23
 \end{aligned}$$

Again some misconceptions here (e.g., the expected value of the max is not the max of the expected values, or their mean). Again, if you made the problem too easy in part (a), you only got a bit of partial credit for just stating the expected value of the beans was 6; that was written down in the problem statement to begin with!